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CS 4379G.251

Urban Air Quality Explorer

As humanity has evolved, there has been an increase of not only urbanization but also industrialization, which has led to changes in our planet's air quality as a result. Air quality is vital because of its strong contributing factor to a person's health and therefore their potential quality of life. The goal of this project is to analyze air levels for PM2.5, O3, NO2, and SO2 pollutants across various cities in the United States from data provided by the EPA and the OpenAQ API. By looking at trends in the air quality index (AQI) and the trends of pollutant concentration themselves. Using this information, I would hope to answer the analytical question of "How do air pollutant levels for PM2.5, O3, NO2, and SO2 pollutants vary across cities and regions in the United States?" by looking at potential temporal, geographic, long-term, and seasonal trends through an interactive dashboard.

Data Sets:

- https://aqs.epa.gov/aqsweb/airdata/download_files.html
- <https://openaq.org>

Proof of accessed data:

Final_Project_data	Today at 10:41 PM	--	Folder
> Daily_NO2	Today at 10:42 PM	--	Folder
> Daily_O3	Today at 10:42 PM	--	Folder
> Daily_PM2.5	Today at 10:37 PM	--	Folder
> daily_SO2	Today at 10:30 PM	--	Folder
annual_aqi_by_cbsa_2019.zip	Today at 10:24 PM	14 KB	ZIP archive
annual_aqi_by_cbsa_2020.zip	Today at 10:24 PM	15 KB	ZIP archive
annual_aqi_by_cbsa_2021.zip	Today at 10:24 PM	15 KB	ZIP archive
annual_aqi_by_cbsa_2022.zip	Today at 10:24 PM	14 KB	ZIP archive
annual_aqi_by_cbsa_2023.zip	Today at 10:24 PM	14 KB	ZIP archive
annual_aqi_by_cbsa_2024.zip	Today at 10:24 PM	14 KB	ZIP archive
annual_aqi_by_cbsa_2025 (1).zip	Today at 10:24 PM	14 KB	ZIP archive
annual_aqi_by_cbsa_2019.csv	Nov 25, 2025 at 7:34 AM	39 KB	CSV Document
annual_aqi_by_cbsa_2020.csv	Nov 25, 2025 at 7:30 AM	38 KB	CSV Document
annual_aqi_by_cbsa_2021.csv	Nov 25, 2025 at 7:26 AM	38 KB	CSV Document

